PROJECT REVIEW – 2

Multimodal Emotion Recognition using Hybrid Deep Learning Models on Facial, Speech, Text and Physiological Data

Table 1: Team members

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3. Dataset Description and Sample Data

The project recognises human emotion from four modalities – facial images (with video as an extension), speech audio, written text and physiological signals – so one or more public benchmark datasets were collected for every modality. Table 2 lists them. For each dataset the following subsections give the download link, a description (features, class label, size), sample data (2 samples per class), a class-imbalance check with the balanced/imbalanced size table and the corresponding figures.

Table 2: Overview of datasets

Modality	Dataset	Size	Class labels	Role in project
Face (image)	FER-2013	35,887 grayscale images, 48×48	7 emotions	Primary – all 5 face models
Face (image)	CK+ (CK+48)	981 grayscale images, 48×48	7 (incl. contempt)	Supplementary
Face (labels)	FER+	35,887 crowd-sourced label votes	8 + unknown	Cleaner labels for FER-2013
Video	RAVDESS (video-speech)	2,160 .mp4 clips (18 of 24 actors downloaded)	8 emotions	Extension
Speech	RAVDESS (audio-speech)	1,440 .wav clips, 24 actors	8 (7 used)	Audio models
Speech	TESS	5,600 .wav clips, 2 speakers	7 emotions	Audio models
Speech	SAVEE	480 .wav clips, 4 speakers	7 emotions	Audio models
Text	GoEmotions	54,263 Reddit comments	27 + neutral → 7 (Ekman)	Text models
Text	dair-ai Emotion	20,000 English tweets	6 emotions	Text models
Text	TweetEval – emotion	5,052 tweets	4 emotions	Supplementary
Physiological	WESAD	15 subjects, 8 chest channels @ 700 Hz	3 affective states	Physiological models

3.1 Facial Expression Datasets

3.1.1 FER-2013

- **Link:** https://www.kaggle.com/datasets/msambare/fer2013 (original: ICML 2013 Challenges in Representation Learning)

Description: FER-2013 contains 35,887 grayscale face images collected automatically with Google image search. Every image is 48 × 48 pixels, the face is roughly centred and occupies most of the frame. The dataset is split into 28,709 training images and 7,178 test images.

- **Features:** the 48 × 48 = 2,304 pixel intensities (0–255, one channel). For the deep models the image is resized to 224 × 224 and replicated to 3 channels; the learned 512–2,048-dimensional embedding of each model is exported as a feature CSV (f0 … fN) for the SMOTE / k-fold protocol.
- **Class label:** 7 emotions – angry, disgust, fear, happy, neutral, sad, surprise.
- **Total number of images:** 35,887 (train 28,709 / test 7,178).

Table 3: FER-2013 class distribution

Class label	Train images	Test images	Total
angry	3,995	958	4,953
disgust	436	111	547
fear	4,097	1,024	5,121
happy	7,215	1,774	8,989
neutral	4,965	1,233	6,198
sad	4,830	1,247	6,077
surprise	3,171	831	4,002
Total	28,709	7,178	35,887



Figure 1: FER-2013 sample images (2 per class)

Sample data: the figure above shows two randomly chosen training images of each class. The images vary strongly in pose, illumination, age, occlusion (hands, glasses) and image quality, which makes FER-2013 a hard, realistic ('in the wild') benchmark.

Table 4: FER-2013 (training set) – imbalanced vs balanced size

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
angry	3,995	13.9%	7,215	+3,220
disgust	436	1.5%	7,215	+6,779
fear	4,097	14.3%	7,215	+3,118
happy	7,215	25.1%	7,215	+0
neutral	4,965	17.3%	7,215	+2,250
sad	4,830	16.8%	7,215	+2,385
surprise	3,171	11.0%	7,215	+4,044
Total	28,709	100%	50,505	+21,796

Imbalance ratio (largest class ÷ smallest class) = 16.5 → the dataset is imbalanced. SMOTE is therefore applied to the training portion so that every class reaches the size of the majority class.

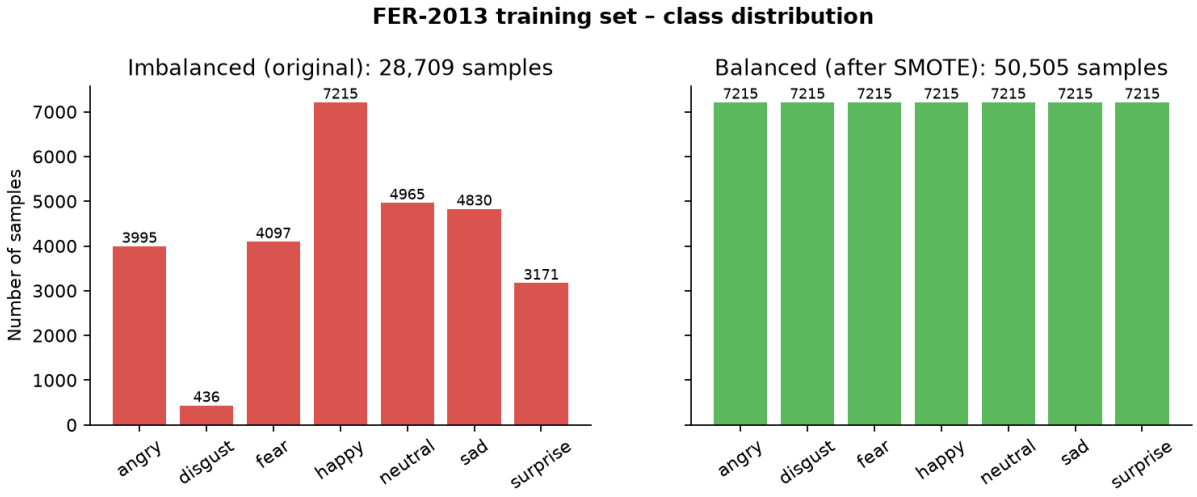


Figure 2: FER-2013 class distribution before and after SMOTE

The 'disgust' class has only 436 training images (1.5%) against 7,215 'happy' images (25.1%), so the dataset is strongly imbalanced. Because SMOTE interpolates between feature vectors, it is applied to the exported deep-feature CSV (not to raw pixels) and only to the training part of every split.

3.1.2 CK+ (Extended Cohn-Kanade, CK+48 version)

- **Link:** https://www.kaggle.com/datasets/shawon10/ckplus (original: Lucey et al., CVPR Workshops 2010)

Description: CK+ is a laboratory dataset of posed facial expressions recorded from 123 subjects. The CK+48 version contains the last (peak-expression) frames of each sequence, cropped to the face and resized to 48 × 48 grayscale – 981 images in total.

- **Features:** 48 × 48 grayscale pixel values (same format as FER-2013).
- **Class label:** 7 classes – anger, contempt, disgust, fear, happy, sadness, surprise.
- **Total number of images:** 981.

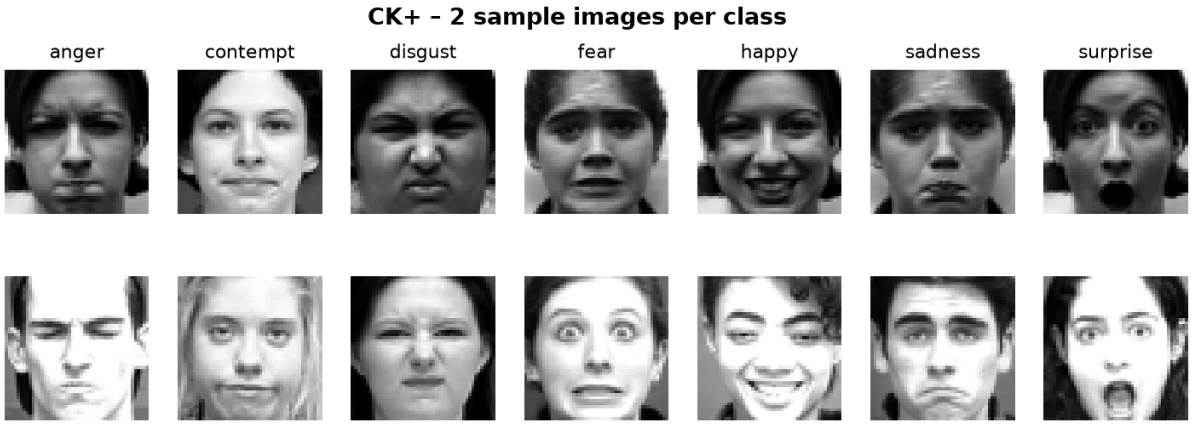


Figure 3: CK+ sample images (2 per class)

Table 5: CK+ – imbalanced vs balanced size

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
anger	135	13.8%	249	+114
contempt	54	5.5%	249	+195
disgust	177	18.0%	249	+72
fear	75	7.6%	249	+174
happy	207	21.1%	249	+42
sadness	84	8.6%	249	+165
surprise	249	25.4%	249	+0
Total	981	100%	1,743	+762

Imbalance ratio (largest class ÷ smallest class) = 4.6 → the dataset is imbalanced. SMOTE is therefore applied to the training portion so that every class reaches the size of the majority class.

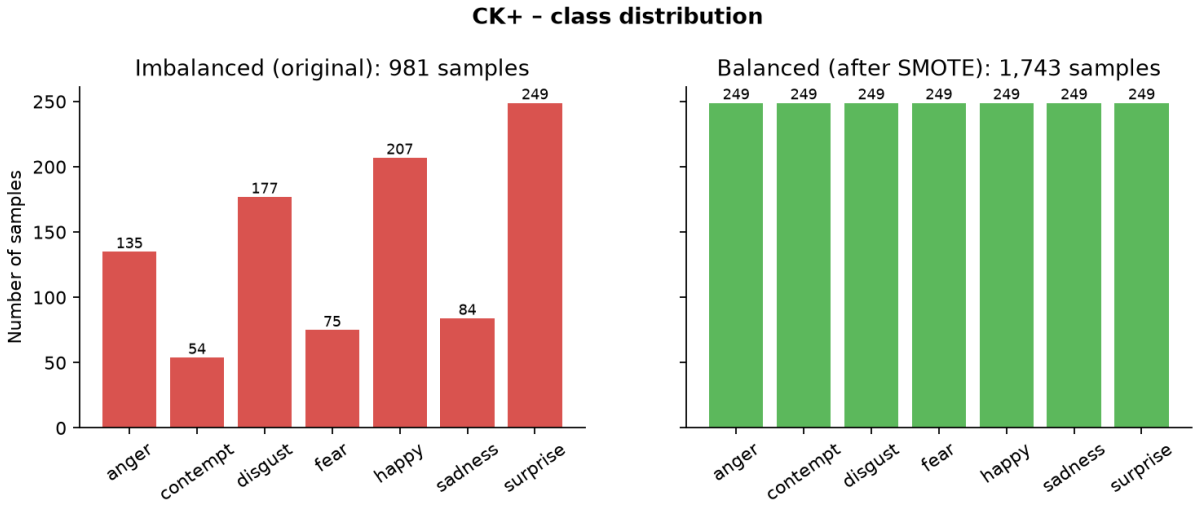


Figure 4: CK+ class distribution before and after SMOTE

3.1.3 FER+ (FER-2013 re-labelled by Microsoft)

- **Link:** https://github.com/microsoft/FERPlus

Description: FER+ does not contain new images; it provides new labels for every FER-2013 image. Ten crowd-workers voted for each image, which removes much of the label noise of the original dataset. It is used as a cleaner-label variant of FER-2013.

- **Features:** the FER-2013 image (referenced by file name) + 10 vote columns.
- **Class label:** neutral, happiness, surprise, sadness, anger, disgust, fear, contempt, plus 'unknown' and 'not a face (NF)'; the majority vote is the label.
- **Total number of rows:** 35,887.

Table 6: FER+ sample rows (CSV) – 2 per class

Usage	Image	neu	hap	sur	sad	ang	dis	fea	con	Label
Training	fer0000000.png	4	0	0	1	3	2	0	0	neutral
Training	fer0000001.png	6	0	1	1	0	0	0	0	neutral
Training	fer0000007.png	0	10	0	0	0	0	0	0	happiness
Training	fer0000008.png	0	10	0	0	0	0	0	0	happiness
Training	fer0000009.png	0	0	6	0	0	0	4	0	surprise
Training	fer0000015.png	0	0	6	0	1	0	3	0	surprise
Training	fer0000006.png	2	0	0	8	0	0	0	0	sadness
Training	fer0000018.png	1	0	2	4	2	0	0	0	sadness
Training	fer0000010.png	2	0	0	0	8	0	0	0	anger
Training	fer0000022.png	1	0	1	1	7	0	0	0	anger
Training	fer0000298.png	1	0	0	1	2	3	1	1	disgust
Training	fer0000480.png	0	0	0	0	2	5	0	1	disgust
Training	fer0000021.png	1	0	1	2	0	0	5	0	fear
Training	fer0000051.png	1	0	3	1	0	0	5	0	fear
Training	fer0000219.png	3	3	0	0	0	0	0	4	contempt
Training	fer0000495.png	3	0	0	1	0	1	0	5	contempt

Table 7: FER+ – majority-vote class sizes

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
neutral	12,906	36.0%	12,906	+0
happiness	9,355	26.1%	12,906	+3,551
surprise	4,462	12.4%	12,906	+8,444
sadness	4,371	12.2%	12,906	+8,535
anger	3,111	8.7%	12,906	+9,795
disgust	248	0.7%	12,906	+12,658
fear	819	2.3%	12,906	+12,087
contempt	216	0.6%	12,906	+12,690
unknown	222	0.6%	12,906	+12,684
NF	177	0.5%	12,906	+12,729
Total	35,887	100%	129,060	+93,173

Imbalance ratio (largest class ÷ smallest class) = 72.9 → the dataset is imbalanced. SMOTE is therefore applied to the training portion so that every class reaches the size of the majority class.

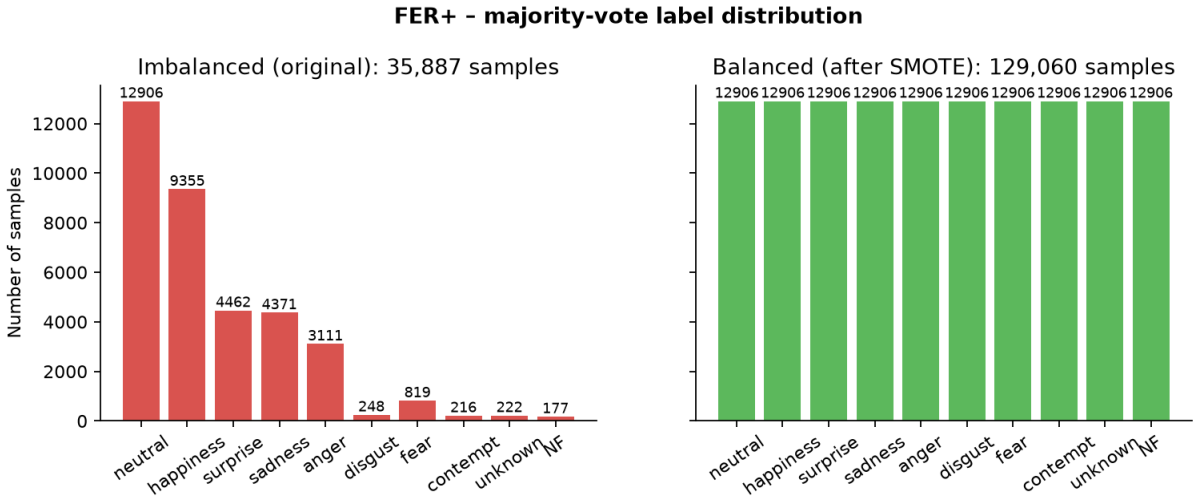


Figure 5: FER+ class distribution before and after SMOTE

3.2 Video Dataset – RAVDESS (Video-Speech)

- **Link:** https://zenodo.org/records/1188976 (Livingstone & Russo, PLoS ONE 2018)

Description: The Ryerson Audio-Visual Database of Emotional Speech and Song contains 24 professional actors (12 female, 12 male) speaking two lexically-matched sentences ('Kids are talking by the door', 'Dogs are sitting by the door') in 8 emotions at two intensities. Each clip is 1280 × 720, 30 fps, about 3–4 s long, provided as audio-video and video-only. Actors 1–18 (18 of 24) have been downloaded so far (2,160 clips); the remaining actors can be added from the same link. Each actor contributes 120 clips (60 audio-video + 60 video-only).

- **Features:** RGB video frames (face region) + synchronised speech; the file name encodes modality, channel, emotion, intensity, statement, repetition and actor.
- **Class label:** neutral, calm, happy, sad, angry, fearful, disgust, surprised.
- **Total number of clips:** 2,160 downloaded (2,880 for all 24 actors).

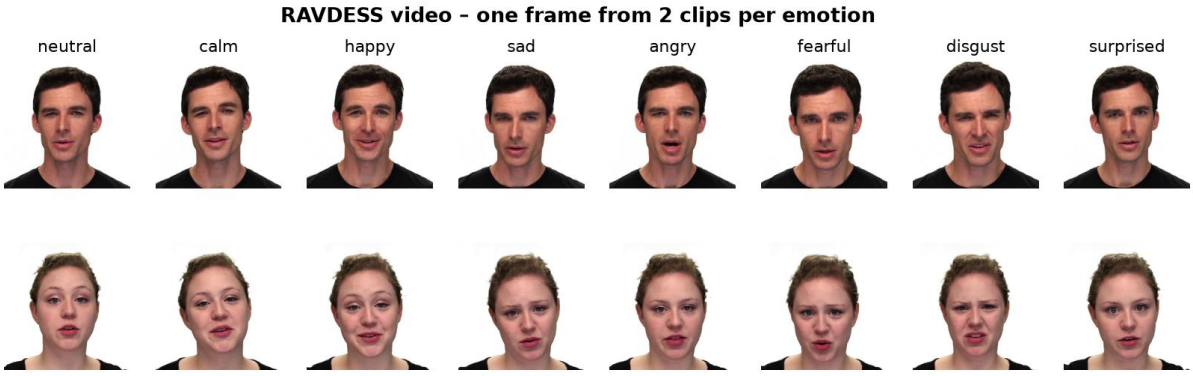


Figure 6: RAVDESS video sample frames (2 clips per emotion)

Table 8: RAVDESS video (downloaded actors) – class sizes

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
neutral	144	6.7%	288	+144
calm	288	13.3%	288	+0
happy	288	13.3%	288	+0
sad	288	13.3%	288	+0
angry	288	13.3%	288	+0
fearful	288	13.3%	288	+0
disgust	288	13.3%	288	+0
surprised	288	13.3%	288	+0
Total	2,160	100%	2,304	+144

Imbalance ratio (largest class ÷ smallest class) = 2.0 → the dataset is imbalanced. SMOTE is therefore applied to the training portion so that every class reaches the size of the majority class.

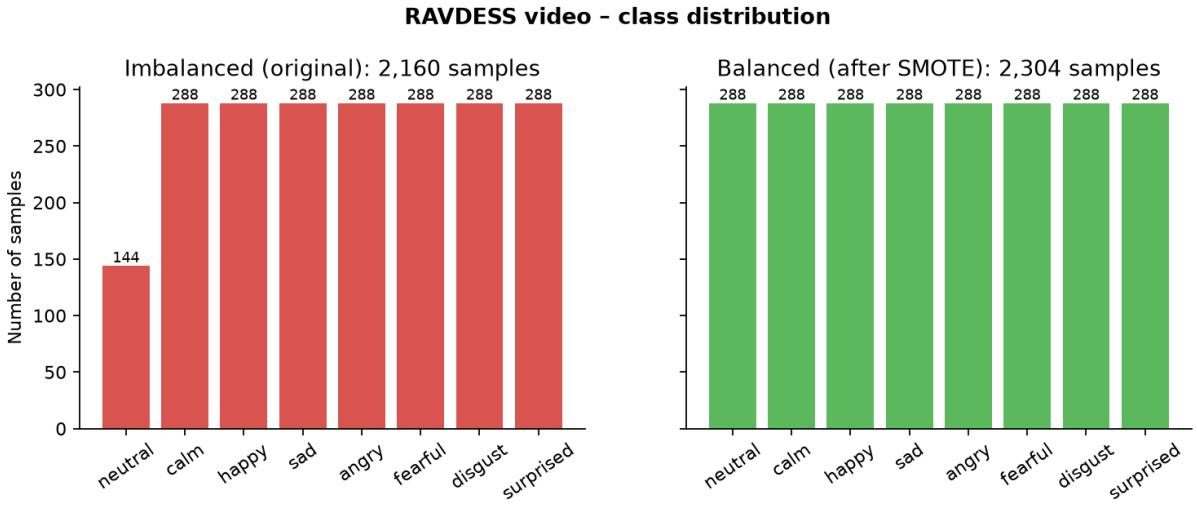


Figure 7: RAVDESS video class distribution before and after SMOTE

RAVDESS is balanced by design except for 'neutral', which has no strong-intensity version and therefore half as many clips as every other emotion.

3.3 Speech Emotion Datasets

Three acted speech-emotion corpora are combined and mapped to the same 7 emotion classes used for the face model, giving one speech dataset of 7,328 clips (RAVDESS 1,248 + TESS 5,600 + SAVEE 480). All clips are converted to mono, resampled to 16 kHz and centre-cropped or zero-padded to 4 s.

Table 9: Speech datasets

Dataset	Link	Speakers	Clips used	Original labels → project labels
RAVDESS (audio-speech)	https://zenodo.org/records/1188976	24 actors (12F/12M)	1,248	8 emotions; 'calm' dropped (no counterpart), others mapped 1:1
TESS	https://tspace.library.utoronto.ca/handle/1807/24487	2 actresses (26 & 64 yrs)	5,600	7 emotions; 'pleasant surprise' → surprise
SAVEE	http://kahlan.eps.surrey.ac.uk/savee/	4 male speakers	480	7 emotions (a, d, f, h, n, sa, su)
CREMA-D (not used yet)	https://github.com/CheyneyComputerScience/CREMA-D	91 actors	0	Downloaded copy contained Git-LFS pointer files only; to be re-downloaded

- **Features:** raw waveform (64,000 samples at 16 kHz) → 64-band log-mel spectrogram for the CNN models, or frame embeddings (768-d) from the pretrained wav2vec 2.0 / HuBERT / AST encoders.
- **Class label:** 7 emotions – angry, disgust, fear, happy, neutral, sad, surprise.
- **Total number of clips:** 7,328.

Table 10: Speech sample data – 2 clips per class

Class label	Dataset	File	Sample rate	Duration
angry	RAVDESS	03-01-05-01-01-01-01.wav	48000 Hz	3.87 s
angry	SAVEE	DC_a01.wav	44100 Hz	3.65 s
disgust	RAVDESS	03-01-07-01-01-01-01.wav	48000 Hz	3.87 s
disgust	SAVEE	DC_d01.wav	44100 Hz	4.36 s
fear	RAVDESS	03-01-06-01-01-01-01.wav	48000 Hz	3.67 s
fear	SAVEE	DC_f01.wav	44100 Hz	3.31 s
happy	RAVDESS	03-01-03-01-01-01-01.wav	48000 Hz	3.47 s
happy	SAVEE	DC_h01.wav	44100 Hz	3.84 s
neutral	RAVDESS	03-01-01-01-01-01-01.wav	48000 Hz	3.30 s
neutral	SAVEE	DC_n01.wav	44100 Hz	3.77 s
sad	RAVDESS	03-01-04-01-01-01-01.wav	48000 Hz	3.84 s
sad	SAVEE	DC_sa01.wav	44100 Hz	4.32 s
surprise	RAVDESS	03-01-08-01-01-01-01.wav	48000 Hz	3.40 s
surprise	SAVEE	DC_su01.wav	44100 Hz	3.25 s

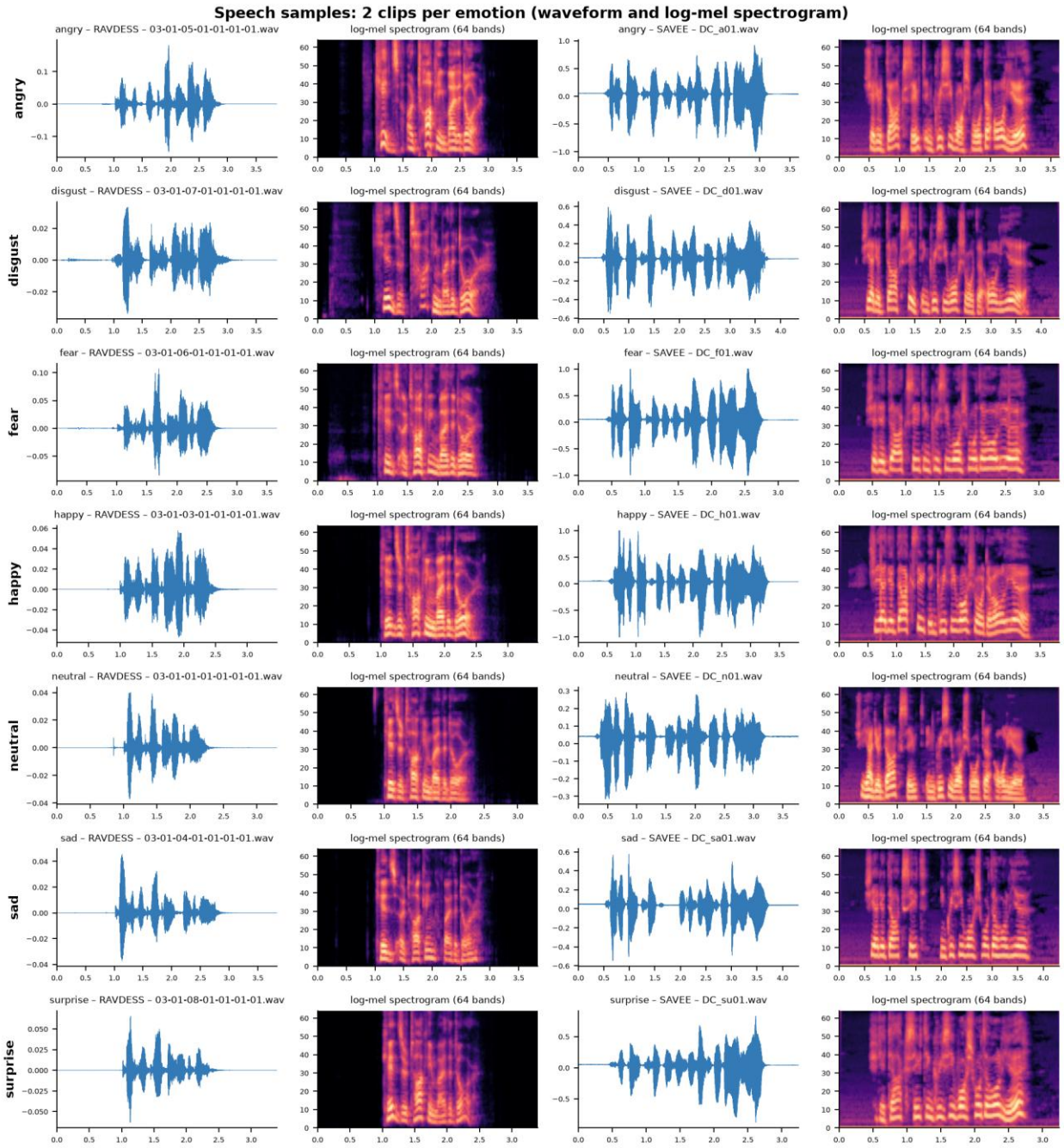


Figure 8: Speech samples: waveform and log-mel spectrogram, 2 clips per class

Table 11: Combined speech dataset – imbalanced vs balanced size

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
angry	1,052	14.4%	1,052	+0
disgust	1,052	14.4%	1,052	+0
fear	1,052	14.4%	1,052	+0
happy	1,052	14.4%	1,052	+0
neutral	1,016	13.9%	1,052	+36

sad	1,052	14.4%	1,052	+0
surprise	1,052	14.4%	1,052	+0
Total	7,328	100%	7,364	+36

Imbalance ratio (largest class ÷ smallest class) = 1.0 → the dataset is approximately balanced. SMOTE is still evaluated (as required by the protocol) but changes the class sizes only slightly.

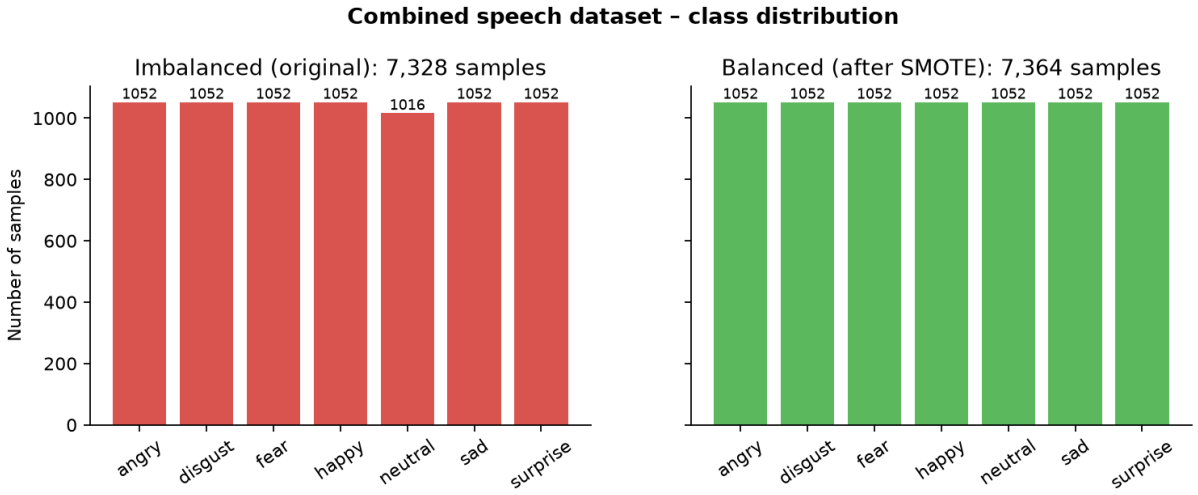


Figure 9: Speech class distribution before and after SMOTE

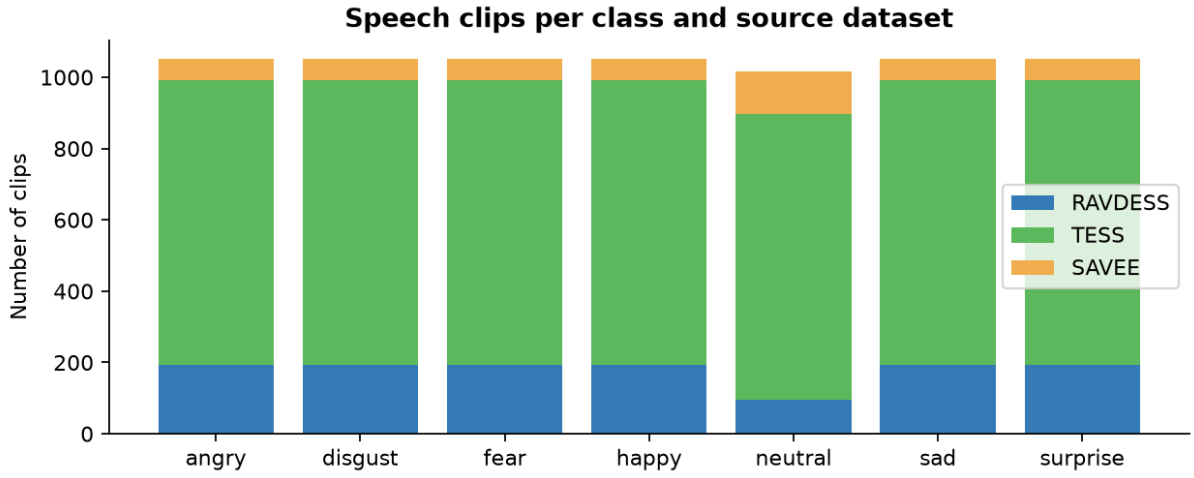


Figure 10: Contribution of each speech corpus to every class

3.4 Text Emotion Datasets

3.4.1 GoEmotions

- **Link:** https://huggingface.co/datasets/google-research-datasets/go_emotions (Demszky et al., ACL 2020)

Description: GoEmotions contains 54,263 English Reddit comments annotated with 27 fine-grained emotions plus neutral (multi-label). Following the authors' Ekman grouping, the 28 labels are merged into the project's 7 classes; comments whose labels fall into more than one group are removed, leaving 49,469 single-label comments (train 39,555 / validation 4,946 / test 4,968).

- **Features:** the comment text (tokenised to at most 64 sub-word tokens); columns in the file: text, labels (list of label ids), id.
- **Class label:** angry, disgust, fear, happy, neutral, sad, surprise (Ekman grouping).
- **Total number of rows:** 54,263 raw; 49,469 after mapping.

Table 12: GoEmotions sample data – 2 rows per class

Class label	Text (CSV sample)
angry	WHY THE FUCK IS BAYLESS ISOING
angry	[NAME] - same fucking problem, slightly better command of the English language.
disgust	youre a disgusting piece of filth.
disgust	Not surprised, damn that sucks. Concussions are awful.
fear	To make her feel threatened
fear	Your coaching is terrible.... be ready and see how [NAME] uses [NAME]
happy	Yes I heard abt the f bombs! That has to be why. Thanks for your reply:) until then hubby and I will anxiously wait 😝
happy	We need more boards and to create a bit more space for [NAME]. Then we'll be good.
neutral	My favourite food is anything I didn't have to cook myself.
neutral	Now if he does off himself, everyone will think hes having a laugh screwing with people instead of actually dead
sad	i got a bump and a bald spot. i feel dumb <3
sad	Pretty sure I've seen this. He swings away with the harness he is wearing. Still looks painful but I think he lives
surprise	OmG pEyToN iSn'T gOoD eNoUgH tO hElP uS iN tHe PlAyOfFs! Dumbass Broncos fans circa December 2015.
surprise	Demographics? I don't know anybody under 35 who has cable tv.

Table 13: GoEmotions (training set) – imbalanced vs balanced size

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
angry	4,293	10.9%	15,216	+10,923
disgust	498	1.3%	15,216	+14,718
fear	541	1.4%	15,216	+14,675
happy	15,216	38.5%	15,216	+0
neutral	12,823	32.4%	15,216	+2,393
sad	2,326	5.9%	15,216	+12,890
surprise	3,858	9.8%	15,216	+11,358
Total	39,555	100%	106,512	+66,957

Imbalance ratio (largest class ÷ smallest class) = 30.6 → the dataset is imbalanced. SMOTE is therefore applied to the training portion so that every class reaches the size of the majority class.

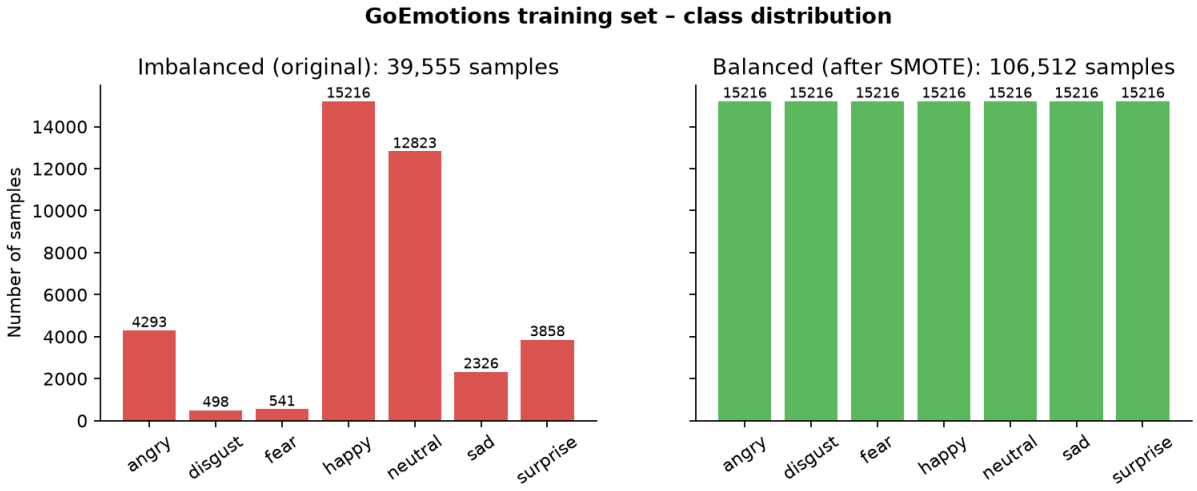


Figure 11: GoEmotions class distribution before and after SMOTE

3.4.2 dair-ai Emotion

- **Link:** https://huggingface.co/datasets/dair-ai/emotion (Saravia et al., EMNLP 2018)

Description: English Twitter messages labelled with six basic emotions using distant supervision (emotion hashtags), split 16,000 / 2,000 / 2,000 for train / validation / test.

- **Features:** text (lower-cased tweet); columns: text, label.
- **Class label:** sadness (0), joy (1), love (2), anger (3), fear (4), surprise (5).
- **Total number of rows:** 20,000.

Table 14: dair-ai Emotion sample data – 2 rows per class

Class label	Text (CSV sample)
sadness	i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and is awake
sadness	ive been feeling a little burdened lately wasnt sure why that was
joy	i have been with petronas for years i feel that petronas has performed well and made a huge profit
joy	i do feel that running is a divine experience and that i can expect to have some type of spiritual encounter
love	i am ever feeling nostalgic about the fireplace i will know that it is still on the property
love	i can t let go of that sad feeling that i want to be accepted here in this first home of mine
anger	im grabbing a minute to post i feel greedy wrong
anger	i think it s the easiest time of year to feel dissatisfied
fear	i feel as confused about life as a teenager or as jaded as a year old man
fear	i now feel compromised and skeptical of the value of every unit of work i put in
surprise	ive been taking or milligrams or times recommended amount and ive fallen asleep a lot faster but i also feel like so funny
surprise	i have seen heard and read over the past couple of days i am left feeling impressed by more than a few companies

Table 15: dair-ai Emotion (training set) – imbalanced vs balanced size

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
sadness	4,666	29.2%	5,362	+696
joy	5,362	33.5%	5,362	+0
love	1,304	8.2%	5,362	+4,058
anger	2,159	13.5%	5,362	+3,203
fear	1,937	12.1%	5,362	+3,425
surprise	572	3.6%	5,362	+4,790
Total	16,000	100%	32,172	+16,172

Imbalance ratio (largest class ÷ smallest class) = 9.4 → the dataset is imbalanced. SMOTE is therefore applied to the training portion so that every class reaches the size of the majority class.

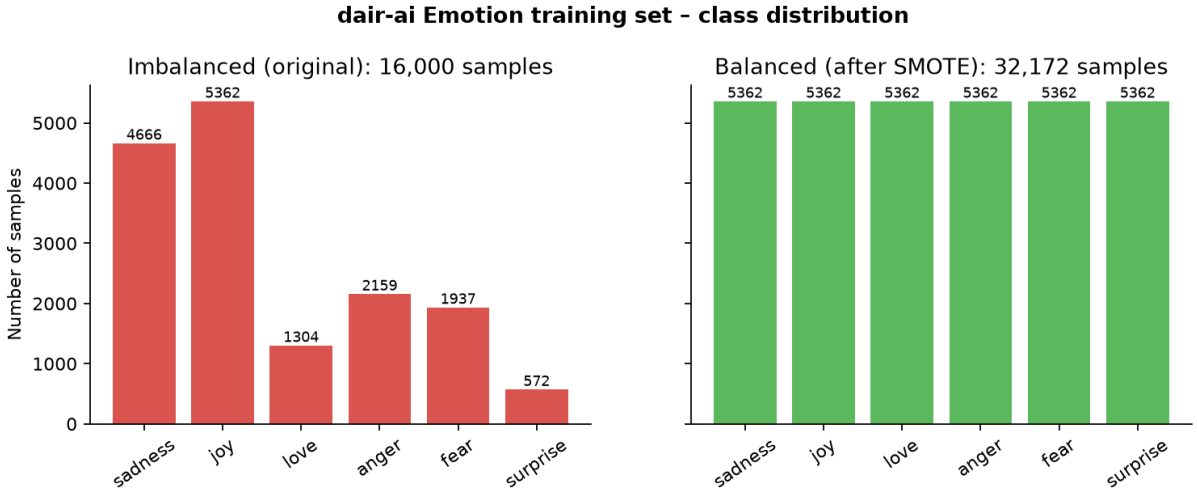


Figure 12: dair-ai Emotion class distribution before and after SMOTE

3.4.3 TweetEval – Emotion subset

- **Link:** https://github.com/cardiffnlp/tweeteval (Barbieri et al., Findings of EMNLP 2020)

Description: The emotion task of the TweetEval benchmark (from SemEval-2018 Task 1) has 5,052 tweets (train 3,257 / validation 374 / test 1,421) labelled with four emotions. It is kept as a supplementary text benchmark.

- **Features:** tweet text (user names replaced by @user); stored as parallel *_text.txt and *_labels.txt files.
- **Class label:** anger (0), joy (1), optimism (2), sadness (3).
- **Total number of rows:** 5,052.

Table 16: TweetEval-emotion sample data – 2 rows per class

Class label	Text (CSV sample)

anger	My roommate: it's okay that we can't spell because we have autocorrect. #terrible #firstworldprobs
anger	Rooneys fucking untouchable isn't he? Been fucking dreadful again, depay has looked decent(ish)tonight
joy	No but that's so cute. Atsu was probably shy about photos before but cherry helped her out uwu
joy	Tiller and breezy should do a collab album. Rapping and singing prolly be fire
optimism	"Worry is a down payment on a problem you may never have'. Joyce Meyer. #motivation #leadership #worry
optimism	I saved him after ordering him to risk his life. I didn't panic but stayed calm and rescued him.
sadness	it's pretty depressing when u hit pan on ur favourite highlighter
sadness	Making that yearly transition from excited and hopeful college returner to sick and exhausted pessimist. #college

Table 17: TweetEval-emotion (training set) – imbalanced vs balanced size

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
anger	1,400	43.0%	1,400	+0
joy	708	21.7%	1,400	+692
optimism	294	9.0%	1,400	+1,106
sadness	855	26.3%	1,400	+545
Total	3,257	100%	5,600	+2,343

Imbalance ratio (largest class ÷ smallest class) = 4.8 → the dataset is imbalanced. SMOTE is therefore applied to the training portion so that every class reaches the size of the majority class.

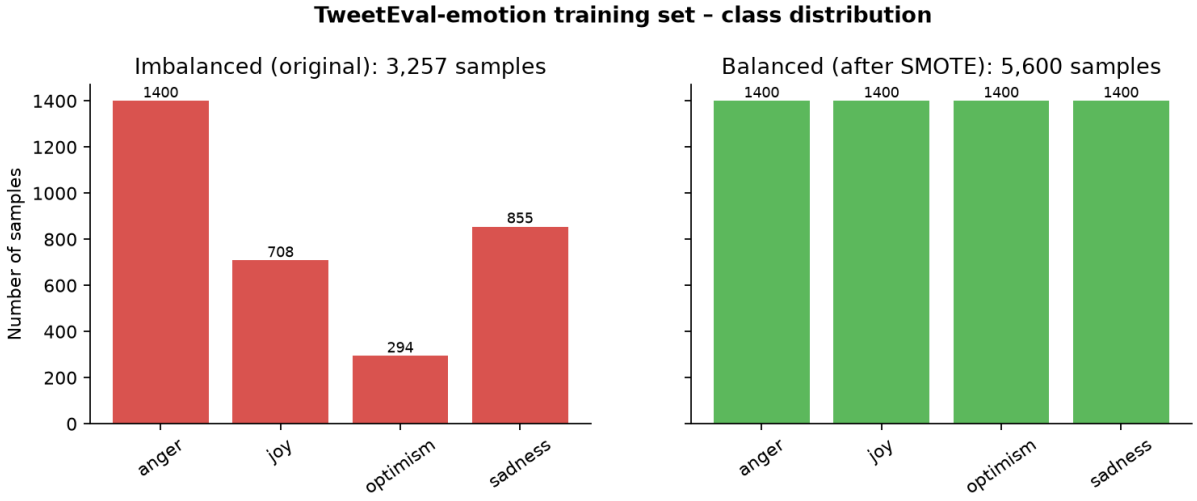


Figure 13: TweetEval-emotion class distribution before and after SMOTE

3.5 Physiological Dataset – WESAD

- **Link:** https://ubi29.informatik.uni-siegen.de/usi/data_wesad.html (also UCI: https://archive.ics.uci.edu/dataset/465/wesad) – Schmidt et al., ICMI 2018

Description: WESAD (Wearable Stress and Affect Detection) records 15 subjects (S2–S17; S1 and S12 discarded by the authors) wearing a chest RespiBAN and a wrist Empatica E4 during a neutral baseline (~20 min), a stress condition (Trier Social Stress Test: public speaking + mental arithmetic, ~10 min) and an amusement condition (funny video clips, ~6.5 min). The chest device samples all channels at 700 Hz.

- **Features:** 8 chest channels – ECG, EDA (electrodermal activity), EMG, respiration, skin temperature and 3-axis acceleration. After preprocessing each sample is a 60-second window of 8 × 2,100 values (35 Hz); for the CSV representation each window is summarised by per-channel statistics (mean, standard deviation, minimum, maximum → 32 features).
- **Class label:** baseline (1), stress (2), amusement (3) – the WESAD protocol labels.
- **Total:** 15 subjects, ~2.25 GB raw; 6,124 windows after segmentation.

Table 18: WESAD – 60 s windows per subject and class

Subject	baseline	stress	amusement	Total
S2	217	112	61	390
S3	217	117	64	398
S4	220	116	63	399
S5	228	118	63	409
S6	225	119	63	407
S7	226	117	63	406
S8	222	123	63	408
S9	225	118	63	406
S10	225	134	63	422
S11	225	125	62	412
S13	225	121	65	411
S14	225	124	63	412
S15	224	126	63	413
S16	225	123	62	410
S17	225	133	63	421
Total	3354	1826	944	6124

Table 19: WESAD feature CSV sample – 2 windows per class (subset of the 32 statistical features; values are per-subject z-scores)

Subj	ECG μ	EDA μ	Resp μ	Temp μ	ECG σ	EDA σ	Resp σ	Temp σ	Label
S5	0.00	-1.55	-0.00	0.25	0.92	0.00	0.68	0.07	baseline
S14	-0.00	-1.57	0.01	-1.68	0.93	0.08	0.70	0.09	baseline
S5	-0.00	0.94	0.09	0.05	1.17	0.05	1.62	0.27	stress
S14	-0.00	0.48	-0.02	0.19	1.25	0.12	1.30	0.02	stress
S5	0.00	-0.81	-0.03	0.87	0.93	0.05	0.53	0.05	amusement
S14	0.00	0.73	0.00	0.60	0.90	0.11	1.44	0.02	amusement

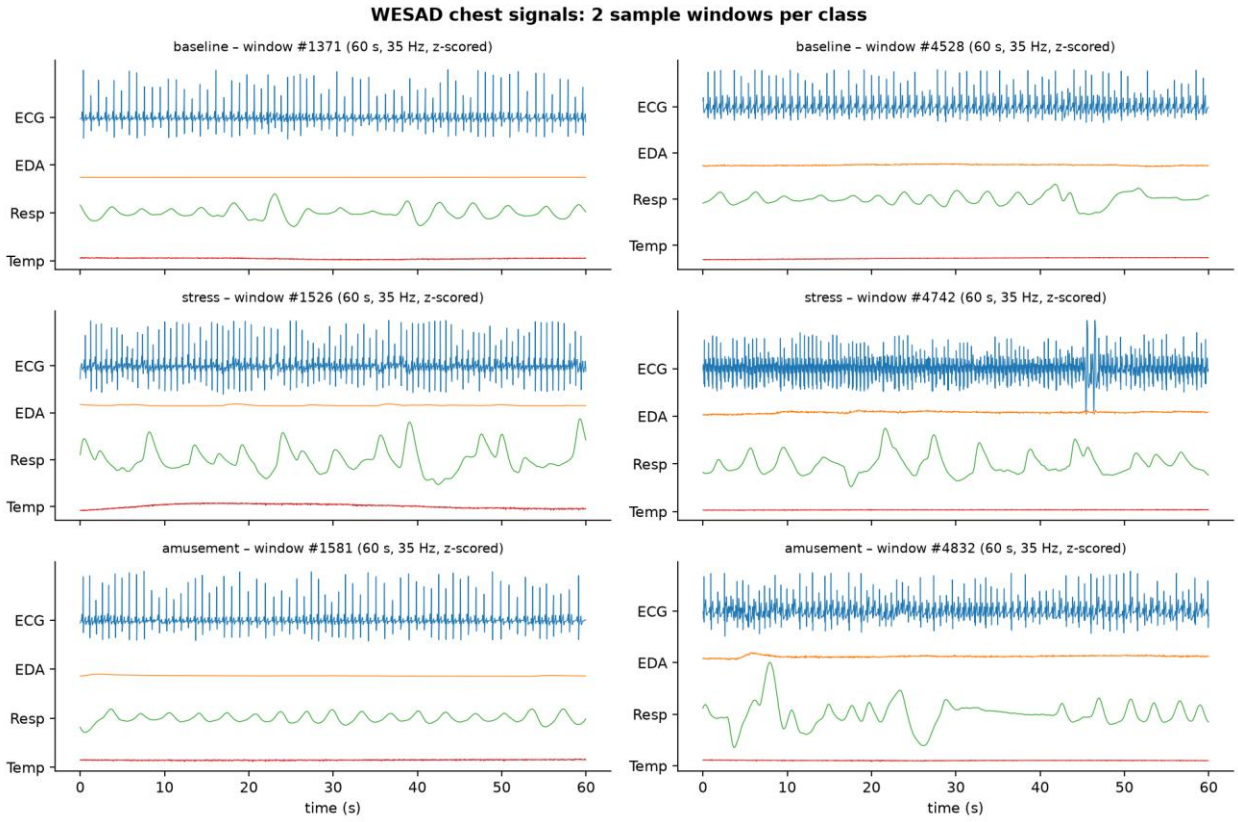


Figure 14: WESAD sample signal windows (2 per class)

Table 20: WESAD windows – imbalanced vs balanced size

Class label	Imbalanced size	Share	Balanced size (SMOTE)	Synthetic samples added
baseline	3,354	54.8%	3,354	+0
stress	1,826	29.8%	3,354	+1,528
amusement	944	15.4%	3,354	+2,410
Total	6,124	100%	10,062	+3,938

Imbalance ratio (largest class ÷ smallest class) = 3.6 → the dataset is imbalanced. SMOTE is therefore applied to the training portion so that every class reaches the size of the majority class.

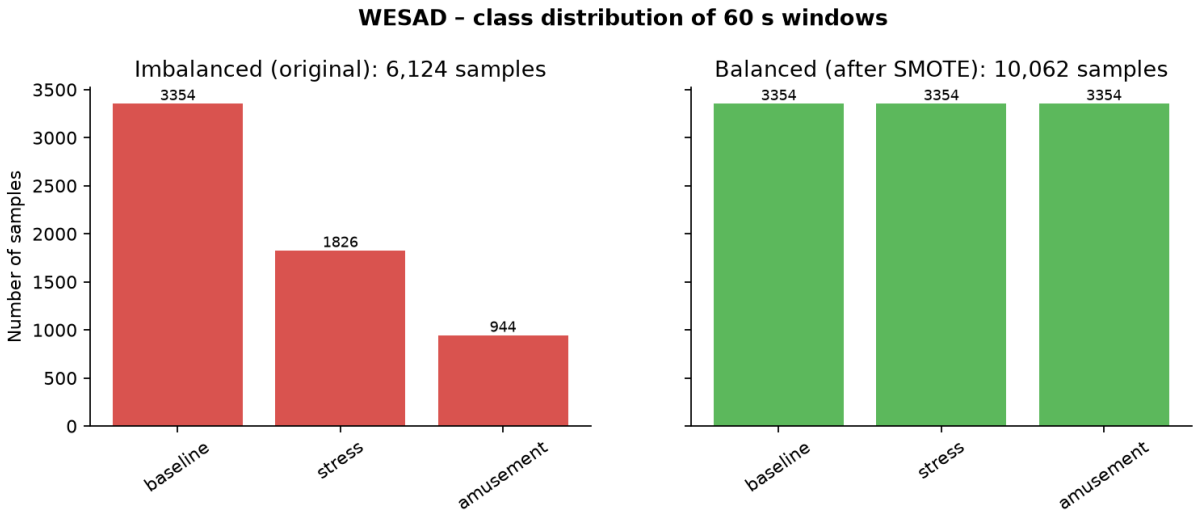


Figure 15: WESAD class distribution before and after SMOTE

4. Methodology

4.1 Steps Involved in the Proposed Methodology

1. Data collection – download the face, video, speech, text and physiological datasets (Section 3).
2. Data loading – read each dataset and convert it to a common tabular form (one row per sample, feature columns + class label).
3. Data preprocessing – modality-specific cleaning, resizing / resampling, normalisation, segmentation, tokenisation and label mapping.
4. Data splitting – (a) hold-out splits of 80:20 and 70:30, and (b) stratified 10-fold cross-validation; the same splits are used for every model of a modality.
5. Class balancing – SMOTE applied to the training data only; every experiment is run both with and without SMOTE.
6. Model building – train five hybrid deep-learning models per modality (20 models in total).
7. Overfitting / underfitting check and rectification – compare training and validation scores; apply early stopping, L1 + L2 regularisation, dropout, data augmentation.
8. Testing – evaluate on held-out data with accuracy, precision, recall, macro-F1, confusion matrix and inference time.
9. Best-model selection – choose the best model per modality and integrate it into the web application.

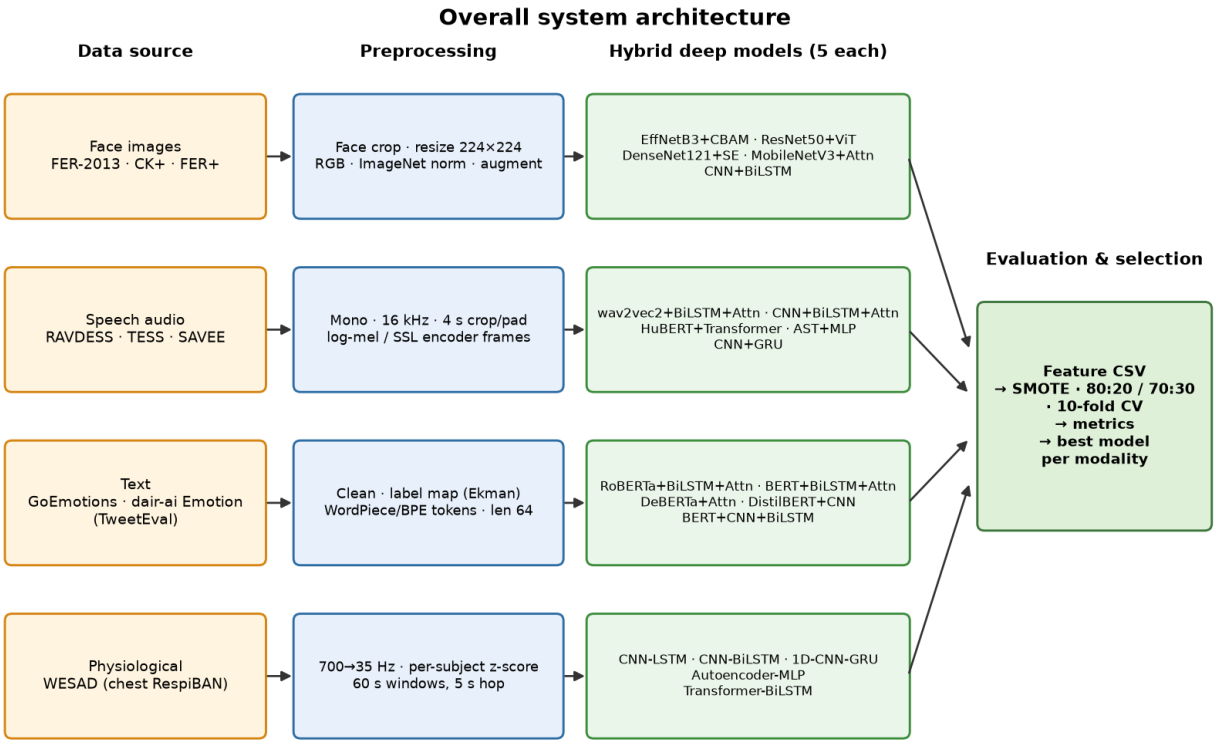


Figure 16: Overall system architecture of the proposed multimodal emotion recognition system

4.2 Detailed Explanation of Each Step

Step 1 – Data collection and loading

All datasets are public research benchmarks (links in Section 3). Images are read from class folders, audio from .wav files whose labels are decoded from the file names, text from Parquet/TXT files and WESAD from the official per-subject pickle files (synchronised signals + labels). Every modality is finally represented as a CSV-style table with numeric feature columns (f0 … fN) and a 'label' column, which is the input format of the evaluation protocol.

Step 2 – Data preprocessing

Table 21: Preprocessing per modality

Modality	Preprocessing operations
Face images	Decode → resize 48×48 to 224×224 → convert grayscale to 3 channels → normalise with ImageNet mean/std. Training-time augmentation: random horizontal flip, ±10° rotation, random resized crop, colour jitter. At inference the face is located with a face detector and cropped first.
Video	Sample frames from each clip → detect and crop the face → process each frame like a face image; the per-frame features form a time sequence.
Speech	Load → convert to mono → resample to 16 kHz → centre-crop / zero-pad to 4 s (64,000 samples) → skip corrupt or placeholder files (< 1 kB) → map labels to 7 classes. CNN models compute a 64-band log-mel spectrogram (1024-point FFT, 20 ms hop) normalised per clip;

	SSL models use the encoder's own feature extractor.
Text	Map labels (GoEmotions 28 → 7 Ekman groups; drop multi-group comments) → tokenise with the encoder's tokenizer (WordPiece / BPE / SentencePiece) → truncate / pad to 64 tokens → attention mask.
Physiological	Take the 8 chest channels at 700 Hz → down-sample to 35 Hz by mean pooling (anti-alias + 20× fewer samples) → z-score per subject (removes person-specific offsets) → keep only labels 1/2/3 → cut 60 s windows with a 5 s hop inside each continuous condition segment.

Step 3 – Data splitting

Two evaluation schemes are used, exactly as in the flowchart: (a) a single stratified hold-out split, run once with 80% training / 20% testing and once with 70% / 30%; (b) stratified 10-fold cross-validation, where the data are cut into 10 equal folds with the same class proportions, each fold is used once as the test set while the other nine train the model, and the mean ± standard deviation of the 10 scores is reported. A fixed random seed makes the splits identical for every model so that the comparison is fair. For WESAD, overlapping neighbouring windows are kept together in the same fold (grouped splitting), and the deep models are trained on subjects different from the test subjects, so no information leaks from training to testing.

Step 4 – Class balancing with SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) creates new minority-class samples instead of duplicating existing ones: for a minority sample x it picks one of its k = 5 nearest neighbours x_nn of the same class and generates $x_{new} = x + \lambda \cdot (x_{nn} - x)$ with λ drawn uniformly from [0, 1]. This is repeated until every class has as many samples as the majority class (the 'balanced size' in the tables of Section 3). SMOTE is fitted on the training part only – never on the test part – otherwise synthetic copies of test samples would inflate the scores. Because SMOTE interpolates numeric vectors, it is applied to the feature CSV (deep embeddings or statistical features), not to raw images, audio or text.

Step 5 – Building the models (hybrid deep learning)

For every modality the five hybrid architectures specified for the project are implemented in PyTorch. Pretrained backbones (ImageNet CNNs, self-supervised speech encoders, pretrained language models) provide strong general features; the attached attention / recurrent / transformer blocks model the emotion-specific spatial or temporal patterns. All models end in a dropout + fully-connected softmax layer. Training uses class-weighted cross-entropy (label smoothing 0.1 for images and speech), the AdamW optimiser with weight decay, a one-cycle (warm-up + cosine decay) learning-rate schedule, mixed-precision (AMP) on an NVIDIA RTX 3050 GPU and gradient clipping.

Table 22: Hybrid models implemented for each modality

Modality	Hybrid model	How it works
Face	EfficientNetB3 + CBAM	EfficientNet-B3 feature map → Convolutional Block Attention Module (channel attention then spatial attention) → global pooling

		→ classifier
Face	ResNet50 + ViT	ResNet-50 (up to layer 3) gives a 14×14 grid of local features → 196 tokens + [CLS] token → 4-layer Vision-Transformer encoder → classifier on [CLS]
Face	DenseNet121 + SE	DenseNet-121 features → Squeeze-and-Excitation channel re-weighting → global pooling → classifier
Face	MobileNetV3 + Attention	Lightweight MobileNetV3-Large → learned spatial-attention pooling → classifier (fast, suitable for real-time use)
Face	CNN + BiLSTM	ResNet-18 CNN → 7×7 feature map read as a 49-step sequence → bidirectional LSTM → attention pooling → classifier (on video the sequence is the frames)
Speech	wav2vec2 + BiLSTM + Attention	wav2vec 2.0 frame embeddings (768-d) → BiLSTM → additive attention pooling → classifier
Speech	CNN + BiLSTM + Attention	Log-mel spectrogram → 4-block 2-D CNN → BiLSTM over time → attention pooling → classifier
Speech	HuBERT + Transformer	HuBERT frame embeddings → 2-layer Transformer encoder → mean pooling → classifier
Speech	AST + MLP	Audio Spectrogram Transformer (AudioSet-pretrained) clip embedding → MLP (LayerNorm, 256 units, GELU, dropout)
Speech	CNN + GRU	Log-mel spectrogram → 2-D CNN → GRU → last hidden state → classifier
Text	RoBERTa + BiLSTM + Attention	RoBERTa-base token states → BiLSTM → attention pooling → classifier (encoder fine-tuned end-to-end)
Text	BERT + BiLSTM + Attention	BERT-base token states → BiLSTM → attention pooling → classifier
Text	DeBERTa + Attention	DeBERTa-v3-base (disentangled attention) → attention pooling → classifier
Text	DistilBERT + CNN	DistilBERT token states → parallel 1-D convolutions (kernel 2/3/4) → max-over-time pooling → classifier
Text	BERT + CNN + BiLSTM	BERT token states → 1-D CNN (local n-grams) → BiLSTM (context) → attention pooling → classifier
Physiological	CNN-LSTM	3-block 1-D CNN (×8 down-sampling) → LSTM → mean pooling → classifier
Physiological	CNN-BiLSTM	Same 1-D CNN → bidirectional LSTM → classifier
Physiological	1D-CNN-GRU	Same 1-D CNN → GRU → classifier
Physiological	Autoencoder-MLP	1-D convolutional autoencoder learns a compressed code (reconstruction loss) → MLP classifies the code; trained jointly
Physiological	Transformer-BiLSTM	1-D CNN tokens + positional embedding → 2-layer Transformer encoder → BiLSTM → classifier

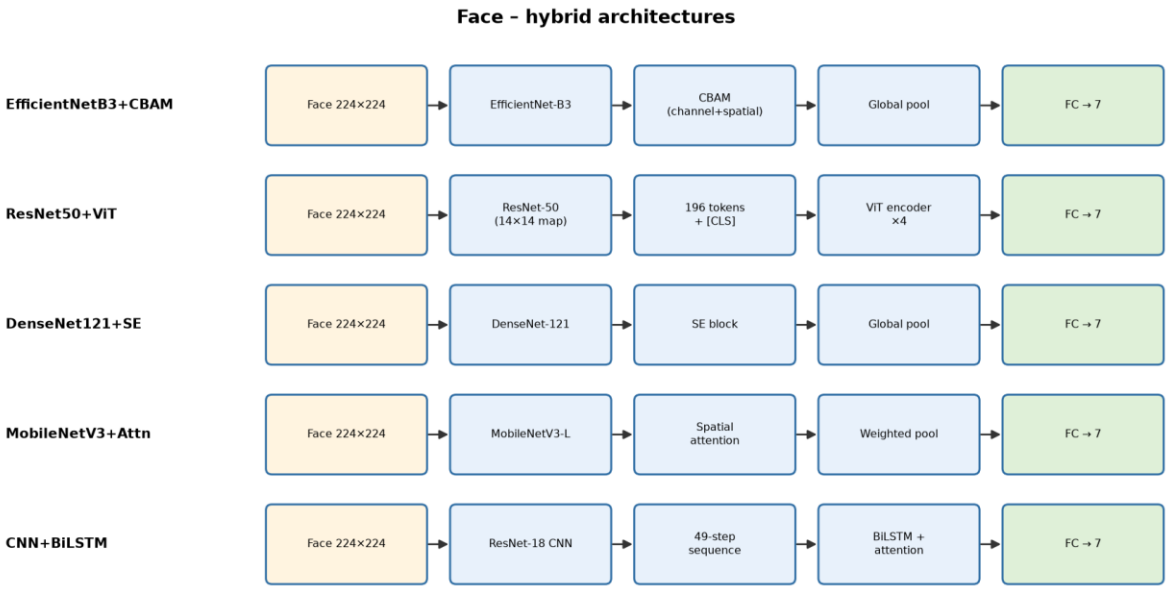


Figure 17: Block diagrams of the five face models

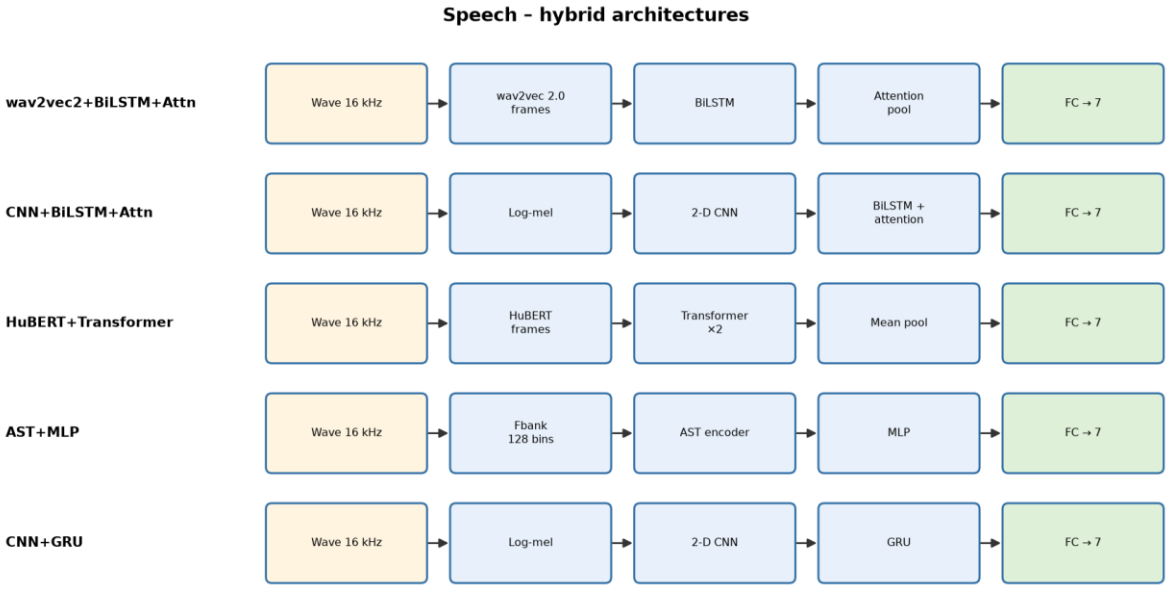


Figure 18: Block diagrams of the five speech models

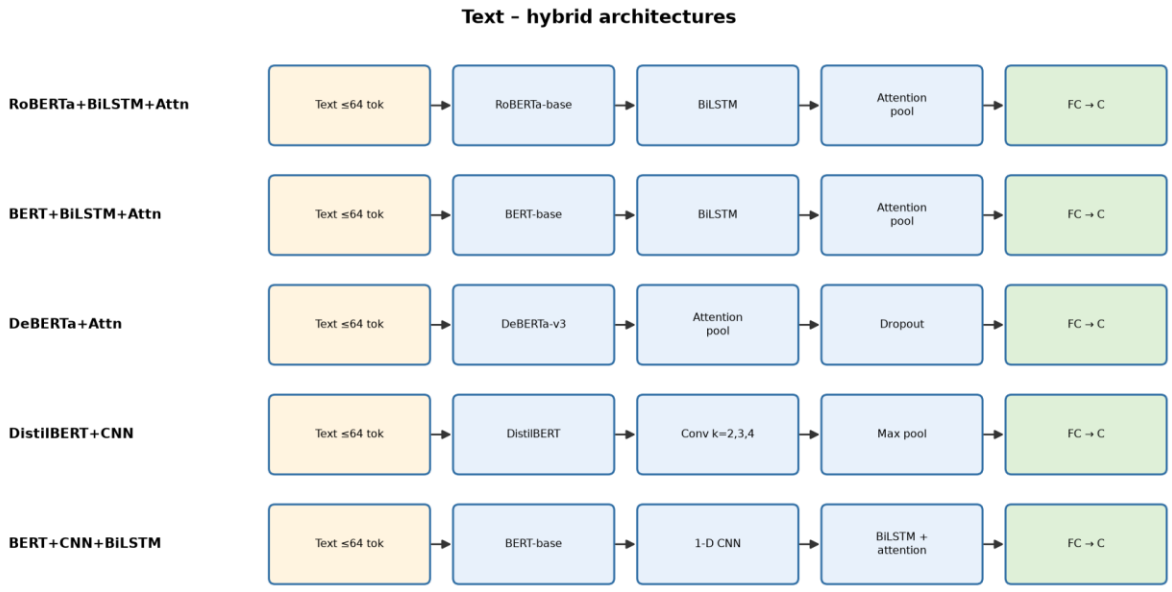


Figure 19: Block diagrams of the five text models (C = 7 for GoEmotions, 6 for dair-ai Emotion)

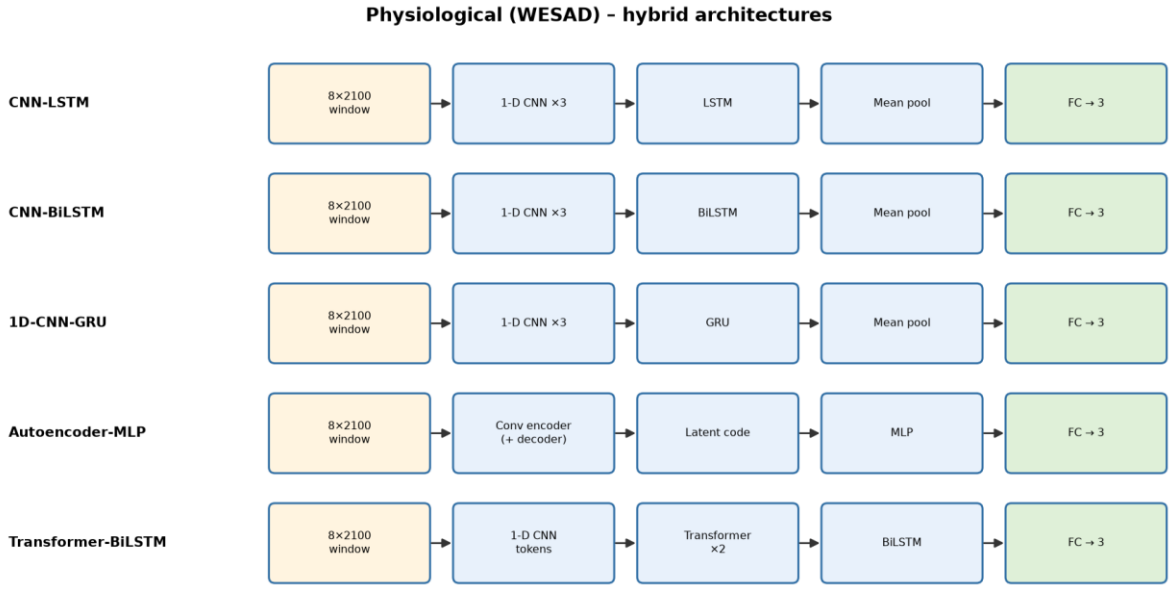


Figure 20: Block diagrams of the five physiological models

Step 6 – Overfitting / underfitting check and rectification

After training, the training accuracy is compared with the validation accuracy. A gap larger than 10 percentage points is flagged as overfitting; low training accuracy (< 50%) is flagged as underfitting. The following remedies are built into training:

- **Early stopping:** a 10% validation slice is carved out of the training data; the model is evaluated after every epoch and the weights with the best validation macro-F1 are kept; training stops after 10 epochs without improvement.
- **L1 and L2 regularisation:** L2 (weight decay 10^{-3}) through the AdamW optimiser and an L1 penalty ($10^{-5} \cdot \Sigma|w|$) added to the loss – together an elastic-net penalty that keeps weights small and sparse.
- **Dropout** (0.2–0.3) before every classifier, label smoothing and data augmentation for images and signals (random jitter noise).
- **Underfitting remedies:** pretrained backbones, more epochs, larger heads or fine-tuning more encoder layers.

Step 7 – Testing the model

Each trained model is evaluated on data it never saw during training. The metrics are accuracy; per-class and macro-averaged precision, recall and F1-score (macro-F1 treats every class equally and is therefore the main metric for imbalanced data); the confusion matrix; and inference time per sample on CPU and GPU (important for the real-time web application).

Table 23: Evaluation metrics

Metric	Formula / meaning
Accuracy	(TP + TN) / (TP + TN + FP + FN) – fraction of correct predictions
Precision	TP / (TP + FP) – of the samples predicted as a class, how many are correct
Recall	TP / (TP + FN) – of the samples of a class, how many are found
F1-score	2 · Precision · Recall / (Precision + Recall); macro-F1 = mean over classes
Confusion matrix	rows = true class, columns = predicted class; shows which emotions are confused
Inference time	milliseconds per sample on CPU and GPU (batch size 1)

Step 8 – Selection of the best model

For every modality the model with the highest mean macro-F1 under 10-fold cross-validation is selected (cross-validation is the most reliable estimate because every sample is tested once); accuracy and inference time are used to break ties. The selected face, speech, text and physiological models are then deployed in the project's web application (FastAPI backend and React frontend), which shows the predicted emotion of each modality in real time.

4.3 Flowchart

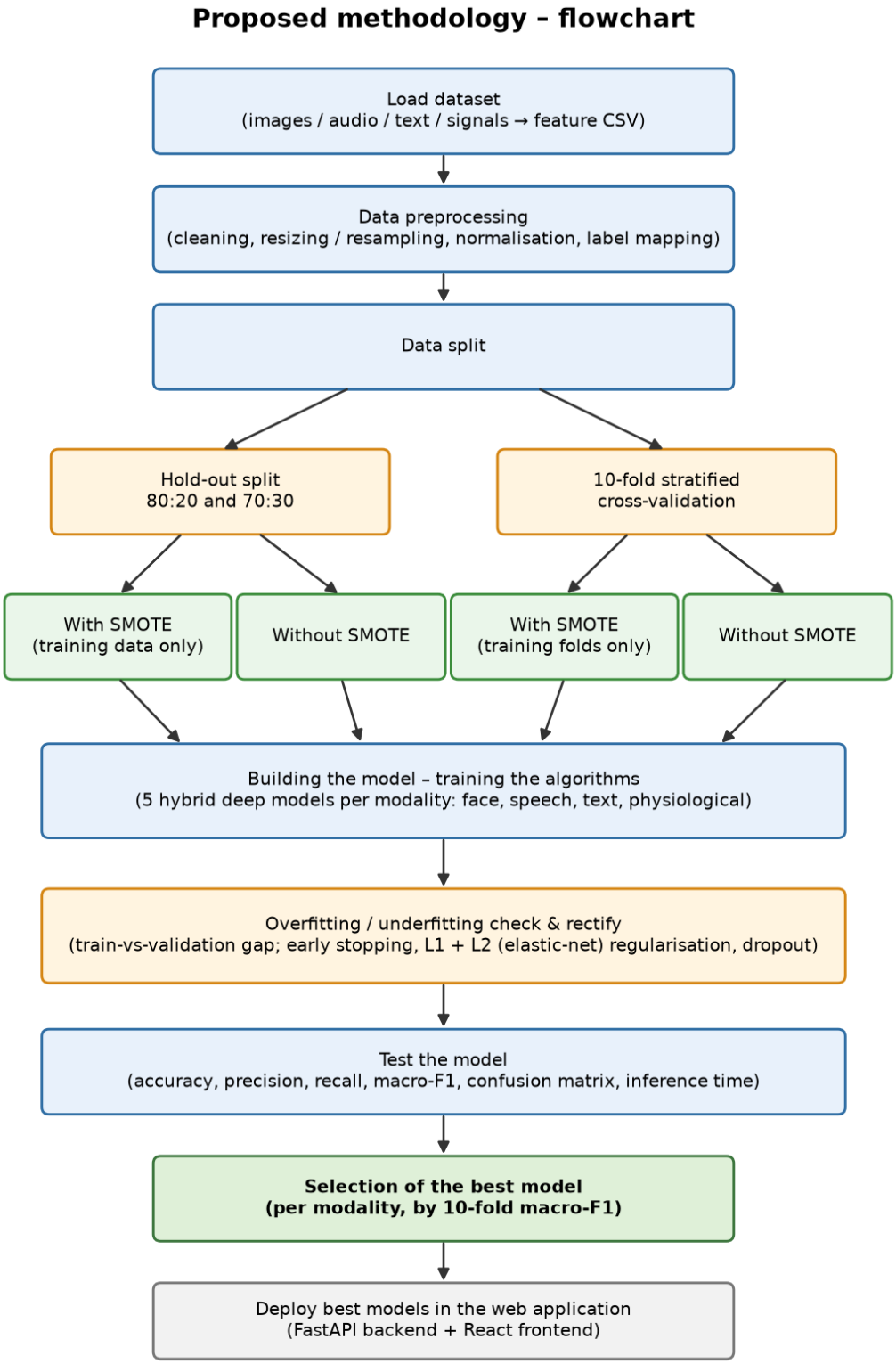


Figure 21: Flowchart of the proposed methodology

The flowchart follows the prescribed pipeline: the dataset (in CSV form) is loaded and preprocessed and then split in two ways. In the left branch a single hold-out split (80:20 or 70:30) is made and the model is trained both with SMOTE-balanced training data and with the original training data. In the right branch 10-fold cross-validation is performed, again with and without SMOTE on the training folds. All variants train the same models, are checked for over/underfitting (early stopping, L1/L2 regularisation), are tested with the same metrics and are finally compared to select the best model.